

Factors Associated with High School Students' Delayed Acquisition of a Driver's License

Insights from Three Northern California Schools

Rodney E. Brown and Susan L. Handy

In recent years the number of teenagers and young adults acquiring a driver's license has declined in many developed countries worldwide, with long-term implications for transportation planning. However, the reasons for this decline are only partially understood. Some research has focused on surveys of young drivers without licenses, but few studies have looked at the characteristics and attitudes of youth of driving age as a whole. Other studies have focused on young adults but not teenagers. This study addressed these gaps by examining the results of a survey of students at three high schools in Northern California. These results indicated that demographic factors such as race-ethnicity and parental education level, which was used as a proxy for socioeconomic status, were correlated with licensing and the age at which a license was obtained. Smartphone use, frequency of participation in after-school activities, and bicycle ownership also corresponded positively with increased licensing rates, while actual bicycle use, enjoyment of riding the bus, and ability to rely on parents for transportation corresponded with decreased licensing rates. Use of electronic communications in general was not found to correspond highly with licensing rates, and environmental concern had no significant relationship. These results suggest the need for further research to separate economic factors from other possible factors influencing licensing rates of high school students.

In recent years a decline in the rate of younger drivers' acquisition of a driver's license has been observed with potential long-term implications for transportation planning. Several studies have documented this trend in developed countries around the world especially for drivers 18 to 29 years old (1–5). Sivak and Schoettle observed this trend in FHWA licensing data (6), although the accuracy of this data has been questioned (7). However, Shults and Williams recently confirmed this trend by analyzing results of an annual survey of high school students. According to their analysis, the proportion of high school seniors with driver's licenses dropped from 85% to 73% from 1996 to 2010, with two-thirds of this decline happening between 2006 and 2010 (8). Recently Tefft et al. reported that only 44% of 18- to 20-year-olds surveyed in 2010 obtained a driver's license within one year of the minimum age (5).

Institute of Transportation Studies, University of California, Davis, 1 Shields Avenue, Davis, CA 95616. Current affiliation for R. E. Brown: Fehr and Peers Transportation Consultants, 2990 Lava Ridge Court, Roseville, CA 95661. Corresponding author: S. L. Handy, slhandy@ucdavis.edu.

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Developing a better understanding of the causes of the decline in the rate of licensing is important for anticipating long-term trends in driving. Though the decline in the rate of licensing of young drivers is apparent, its causes are less clear. Williams's study of potential drivers of ages 15 to 18 identified reasons for delaying licensing including lack of car access, busyness with other activities, cost, and lack of need (9). Tefft et al. found a similar variety of reasons for 18- to 20-year-old survey respondents (5). A few researchers examined graduated driver licensing policies (under which drivers younger than 18 gain driving privileges in stages) implemented in recent years as another possible cause for delay (5, 9, 10), but evidence on the importance of these requirements is mixed. On the basis of this previous research, driver's licensing decisions are likely related to income, access to a car and other modes of travel, and activities and attitudes.

Although previous studies have directly asked young adults about their reasons for delaying acquisition of a driver's license, little of this work has directly explored the relationship of their activities and attitudes to licensing decisions. This paper aimed to address this gap. Analysis of data from a survey of students in three Northern California high schools indicated that sociodemographic factors, extracurricular activities, bicycle ownership and use, and attitudes about transportation options were correlated with licensing and the age at which a license was obtained. The results suggest the need for further research to separate economic factors from other possible factors influencing licensing rates of high school students.

LITERATURE REVIEW

As shown in previous studies, a high school student's choice of whether and when to obtain a driver's license may be related to many factors.

Demographic factors appear to be strongly related to licensing and also to availability of other transportation options. Tefft et al. (5) and Curry et al. (11) reported increasing licensing rates with increasing income. Similarly, Blumenberg et al. found economic factors such as employment status and income to be the biggest factors impacting youth travel, especially after the recent economic downturn (10). Williams (9) and Tefft et al. (5) both reported lower licensing rates for black and Hispanic respondents. Respondents to these surveys and to a survey by Schoettle and Sivak (4) of young adults who delayed licensing reported that they did not need a license to do what they wanted or that they obtained transportation from others. Davis et al. reported that young adults were increasingly bicycling, walking, and using public transit instead of driving (12). Other household factors,

such as whether a child splits time between multiple households or lives with siblings, may also be expected to influence licensing.

Participation in activities also appears to be associated with the decision to obtain a driver's license. Several surveys of youth who delayed licensing reported being too busy as a factor (4, 5, 9). However, the opposite relationship, in which participation in activities before or after school increased the demand for transportation and was thus associated with more driving and earlier licensing, might also be expected.

Some researchers have hypothesized that the increasing use of electronic communications has contributed to the decline in licensing (13); however, no clear conclusion has been reached (14, 15). Respondents to surveys conducted by Williams (9) and Tefft et al. (5) reported online communications as a reason for delay much less frequently than other reasons. Additionally, Le Vine et al. found a positive relationship between online activity and holding a driver's license (16), and Blumenberg et al. found a positive relationship between web use and personal miles traveled (10).

Although a substantial body of research has shown that attitudes are an important influence on travel behavior (17), few studies have so far examined the link between the attitudes of teenagers and their decisions about licensing. Attitudes about the environment do not appear to be strongly related to licensing decisions. Environmental concerns were cited infrequently compared with other reasons for not having a driver's license by respondents to the survey by Schoettle and Sivak (4) and to a survey of unlicensed British people of ages 17 to 29 analyzed by Le Vine et al. (18). Other factors that might be expected to influence driver licensing include perception of different travel modes, attitude about physical activity, and concerns about being rushed in the morning or having lots of things to carry.

METHOD

The analysis in this study was based on data from a cross-sectional, exploratory study originally designed to identify key factors affecting whether or not high school students bicycle to school. Several

questions regarding driver licensing were part of this survey; these questions included the student's driver's license status and, if the student did have a license, when it was obtained. The survey was administered in the spring of 2013 to students at three schools in Northern California: Davis Senior High School (Davis High) in the city of Davis, Sequoia High School (Sequoia High) in Redwood City, and Tamalpais High School (Tamalpais High) in Mill Valley.

Setting

The three schools and the communities in which they are situated differ from each other in important ways, including the sociodemographic makeup of each community, the topography of the catchment area for each high school, and the prevalence of nonautomobile travel modes.

All three communities are somewhat wealthier than the state as a whole, as shown in U.S. Bureau of Census data presented in Table 1. Davis is a relatively wealthy university town; Mill Valley (where Tamalpais High is located), situated in Marin County, is wealthy, and the population is white. However, Redwood City, where Sequoia High is located, is more economically and racially diverse and has more students whose first language is not English than do Davis High or Tamalpais High.

Survey Methods

A printed questionnaire was used to collect data during the first week of May 2013. The content was based on a 2009 survey administered at Davis High (21), including questions about students' transportation to school; cross-streets of their home locations; attitudes about transportation and other issues; and basic demographic information including race-ethnicity, gender, age, and parental education level.

While the survey was intended to be easy for all students to complete on their own in less than 10 min, the survey was difficult for some respondents in a few classes for students whose second language

TABLE 1 Sociodemographic Attributes of Three High School Locations and the State of California

Attribute	Davis (Davis High)	Redwood City (Sequoia High)	Mill Valley (Tamalpais High)	State of California
Population ^a	65,616	77,036	13,931	37,325,068
Family income ^a				
Median (\$)	108,058	90,566	156,534	69,883
<\$25,000 (%)	10.7	8.7	3.1	15.6
≥\$200,000 (%)	16.5	17.3	38.8	8.6
Race or ethnicity ^a (%)				
White (only)	64.2	74.8	89.2	62.3
Black or African American (only)	2.1	2.5	0.9	6.0
Asian (only)	21.9	10.3	5.4	13.2
White alone or with any other race	69.9	78.0	91.2	65.7
Black or African American (alone or with any other race)	3.2	3.2	1.4	7.1
Asian (alone or with any other race)	26.2	12.5	7.6	14.9
Hispanic (of any race)	13.0	39.3	5.6	37.6
Workers commuting by ^a (%)				
Car	65.3	84.8	70.2	84.6
Public transportation	7.2	3.6	9.0	5.1
Walking	3.1	3.1	3.8	2.8
Bicycle	18.6	1.7	2.6	1.0
Students who were English learners ^b (%)	4.1	19.6	1.9	21.6
Students who qualified for free or reduced-price meals ^b (%)	19.1	46.7	9.8	58.0

NOTE: Catchment areas for students attending each high school do not align directly with each city boundary.

^aU.S. Bureau of the Census, 2008–2012 American Community Survey 5-Year Estimates (19).

^bDataquest (20).

was English. In these cases, the teacher helped students by reading the questions aloud to the class.

Survey Response

Responses totaled 3,090, and the overall participation rate was 61.8%. By school, the participation rates were 70.8% at Davis High, 53.8% at Sequoia High, and 62.1% at Tamalpais High [percentages are based on enrollment data from the California Department of Education (21)]. Although all teachers were encouraged to administer the survey, in

some classes in each school the survey was either never administered or not returned.

Sociodemographic characteristics of the students surveyed are presented in Table 2. Though the boundaries of the area served by each school do not align directly with each city and often extend into adjacent jurisdictions, differences in the characteristics of surveyed students at each school correspond to the differences in each community as shown in Table 1. In addition, because it was expected that students would be unable to report family income accurately, they were asked to report parental education level, and this information was used as a proxy for family income.

TABLE 2 Sociodemographic Attributes of Students Surveyed

Characteristic	Overall		Davis		Sequoia		Tamalpais	
	Number	Share (%)	Number	Share (%)	Number	Share (%)	Number	Share (%)
High school	3,076	100.0	1,227	39.9	1,088	35.4	761	24.7
Age group (years)								
≤15.49	409	13.3	26	2.1	223	20.5	160	21.0
15.50–15.99	354	11.5	145	11.8	110	10.1	99	13.0
16.00–16.49	488	15.9	209	17.0	177	16.3	102	13.4
16.50–16.99	438	14.2	202	16.5	139	12.8	97	12.7
17.00–17.49	418	13.6	200	16.3	126	11.6	92	12.1
17.50–17.99	348	11.3	171	13.9	94	8.6	83	10.9
18.00–18.49	401	13.0	199	16.2	121	11.1	81	10.6
≥18.50	90	2.9	36	2.9	36	3.3	18	2.4
How would you describe your race–ethnicity? (Check all that apply.) ^a								
American Indian–Native American	100	3.3	50	4.1	27	2.5	23	3.0
Asian	432	14.0	252	20.5	85	7.8	95	12.5
Black or African American	156	5.1	69	5.6	44	4.0	43	5.7
Hispanic or Latino	887	28.8	207	16.9	605	55.6	75	9.9
Pacific Islander	122	4.0	48	3.9	45	4.1	29	3.8
White or Caucasian	1,745	56.7	761	62.0	400	36.8	584	76.7
Other	168	5.5	78	6.4	42	3.9	48	6.3
What grade are you in?								
9	462	15.0	3	0.2	266	24.4	193	25.4
10	937	30.5	422	34.4	305	28.0	210	27.6
11	873	28.4	408	33.3	273	25.1	192	25.2
12	797	25.9	391	31.9	243	22.3	163	21.4
What is your gender?								
Male	1,462	47.5	591	48.2	501	46.0	370	48.6
Female	1,566	50.9	617	50.3	566	52.0	383	50.3
What is the highest level of education completed by whichever parent or guardian has the most education?								
Some high school	170	5.5	33	2.7	119	10.9	18	2.4
High school	256	8.3	48	3.9	170	15.6	38	5.0
Some college	311	10.1	88	7.2	152	14.0	71	9.3
Associate degree	106	3.4	32	2.6	54	5.0	20	2.6
Bachelor's degree	664	21.6	310	25.3	148	13.6	206	27.1
Advanced degree	1,017	33.1	582	47.4	166	15.3	269	35.3
Other	68	2.2	10	0.8	48	4.4	10	1.3
Do not know	378	12.3	88	7.2	184	16.9	106	13.9
Do you have a cell phone?								
No	168	5.5	55	4.5	90	8.3	23	3.0
Yes, but not a smartphone	688	22.4	309	25.2	233	21.4	146	19.2
Yes, a smartphone with a data plan	2,140	69.6	842	68.6	730	67.1	568	74.6
What is the most recent driver's license or permit you have obtained?								
Provisional license	363	11.8	172	14.0	77	7.1	114	15.0
Regular driver's license	609	19.8	322	26.2	128	11.8	159	20.9
Driver learner's permit	553	18.0	261	21.3	159	14.6	133	17.5
I do not have a license or permit	1,450	47.1	444	36.2	672	61.8	334	43.9

^aRace–ethnicity share totals exceed 100% because students could check more than one response.

Data Analysis

Two dependent variables were analyzed. One was binary, reflecting whether a student had no license or either a driver's license or learner's permit. Only students 15½ years old (the minimum age for a learner's permit in California; 16 is the minimum age for a driver's license) or older were considered in this analysis. A second dependent variable categorized students in their senior year as early drivers (those obtaining a license before age 16½, i.e., during their first six months of eligibility), late drivers (those obtaining a license at or after age 16½, therefore after their first six months of eligibility), or nondrivers (those without a license or with only a learner's permit). Only students in Grade 12 and at least 16½ years old were considered in this analysis.

Explanatory variables included demographic factors, activity and behavior factors, and attitudinal factors. On the basis of previous studies, when data were available for a variable, hypotheses were made about relationships between factors that might be related to if or when a driver's license was obtained as depicted in Table 3.

TABLE 3 Factors Influencing If or When Driver's License Is Obtained

Independent Variable	Impact on Obtaining Driver's License
Demographic factors	
Female	–
Parental education level	+
Split time between different homes of separated parents	+
Distance from home to school	+
Asian	+
Black or African American	–
Hispanic or Latino	–
White or Caucasian	+
Younger siblings living in household	+
Older siblings living in household	–
Activities and behaviors	
After-school activities	+
Arrival before or after scheduled school start time	+
Own cell phone or smartphone	– if enables other modes, + if income related
Use of electronic communications to arrange transportation	–
Bicycle access	– if related to bicycle usage, + if income related
Use of bicycle for other reasons	–
Car transportation to middle school	+
Attitudes	
Rushed to get ready in the morning	+
Lots of stuff to carry	+
Like being physically active	–
Like bicycling	–
Like riding bus	–
Like being driven	+
Consider driving coolest way to get to school	+
Consider bicycling coolest way to get to school	–
Consider traveling with friends important	± depending on friends' modes and distances
Consider protecting environment important	–
Can rely on parents for transportation	–

NOTE: – = negative impact; + = positive impact.

Demographic factors included gender, race–ethnicity, parental education, and family living arrangement. Activities and behaviors included school arrival time, participation in after-school activities, and electronic communications. Attitudinal factors included attitudes about transportation, physical activity, and availability of parents to provide transportation.

Attitudinal factors were measured on a 5-point Likert scale and then evaluated as dummy variables by grouping “agree” and “strongly agree” responses. Students were also asked to report the intersection closest to their home; these data were entered into ArcGIS, and network distances to school were calculated successfully for 72.3% of the respondents.

Bivariate analysis was used to evaluate relationships between these factors and the dependent variables. Regression models were also created for each hypothesized dependent variable. For the model of all students and whether a license was held or not, a binary logistic regression model was estimated. For the model of all high school seniors and early, late, or no licensing, a multinomial logistic model was estimated.

Limitations

There were several limitations to this survey and analysis. First, this survey was originally designed to measure bicycle commuting to high school. A survey designed to study licensing behavior would have likely included more driving-specific topics such as parental licensing history, attitudes about the licensing process, income, and household car availability. Some students may have misunderstood the difference between a provisional learner's permit, a provisional license, and a regular license. Additionally, some students did not take the survey seriously. Although 14 responses that were clearly not serious were removed, responses to some questions that the respondent did not answer honestly were likely included. Lastly, some bias may have been introduced by students whose second language was English and who may have had problems answering the survey.

RESULTS

Of all 2,483 valid responses from students of age 15½ or older, 1,476 or 59.4% had a license or learner's permit and 1,007 or 40.6% did not. These results are slightly different than those from Williams's survey of U.S. teenagers of ages 15 to 18 in November 2010, which reported 63% of respondents with licenses or learner's permits and 37% without (9). Of all 730 valid responses from students in Grade 12 of ages 16½ or older, 216 or 29.6% received their licenses before age 16½, 238 or 32.6% received their licenses at or after age 16½, and 276 or 37.8% had no license or only a learner's permit. This licensing rate is lower than that reported by Shults and Williams who found that 73% of U.S. high school seniors surveyed in 2010 had a license (8).

Bivariate Analysis

Whether a student held a license or not was strongly associated with several demographic factors (Tables 4 and 5). Beyond the control factors of high school, age, and grade, race–ethnicity and parental education level were both associated with differing levels of license

TABLE 4 Students Ages 15½ and Older With and Without a License or Learner's Permit by Demographic Characteristic

Characteristic	Share of Students ^a (%)	n	License or Permit (%)		p-Value ^b
			Have	Do Not Have	
High school					
Davis	46.1	1,144	64.6	35.4	<.0005
Sequoia	31.3	776	44.6	55.4	
Tamalpais	22.7	563	69.4	30.6	
Age group (years)					
15.50–15.99	14.1	350	34.6	65.4	<.0005
16.00–16.49	19.4	482	49.4	50.6	
16.50–16.99	17.0	423	58.4	41.6	
17.00–17.49	16.5	410	64.6	35.4	
17.50–17.99	13.6	338	74.6	25.4	
18.00–18.49	15.8	392	76.8	23.2	
≥18.50	3.5	88	59.1	40.9	
What grade are you in?					
9	3.0	75	24.0	76.0	<.0005
10	33.6	834	45.2	54.8	
11	32.9	818	62.2	37.8	
12	30.4	756	75.7	24.3	
What is your gender?					
Male	47.5	1,180	58.6	41.4	.456
Female	51.1	1,269	60.2	39.8	
What is the highest level of education completed by whichever parent or guardian has the most education?					
Other or do not know	12.6	312	38.5	61.5	<.0005
Less than bachelor's degree	26.1	648	44.6	55.4	
Bachelor's degree or higher	58.9	1,462	70.7	29.3	
Do you live with your parents or guardians in one home, or split your time at different homes of separated parents?					
I live in one place.	85.8	2,131	58.8	41.2	.031
I live at more than one place.	11.7	291	65.6	34.4	
Distance from home to school (GIS calculated) (mi)					
0–0.49	4.0	100	58.0	42.0	.347
0.50–0.99	12.2	303	58.7	41.3	
1.00–1.99	24.9	618	64.2	35.8	
2.00–3.99	24.0	595	64.2	35.8	
≥4.00	10.8	267	61.0	39.0	
How would you describe your race–ethnicity? (Check all that apply.)					
American Indian or Native American	3.4	84	56.0	44.0	.598
(Not checked)	94.6	2,350	59.4	40.6	
Asian	13.7	340	54.4	45.6	.054
(Not checked)	84.3	2,094	60.1	39.9	
Black or African American	5.0	125	45.6	54.4	.002
(Not checked)	93.0	2,309	60.1	39.9	
Hispanic or Latino	28.1	697	38.9	61.1	<.0005
(Not checked)	70.0	1,737	67.5	32.5	
Pacific Islander	3.9	96	46.9	53.1	.015
(Not checked)	94.2	2,338	59.8	40.2	
White or Caucasian	58.8	1,459	72.4	27.6	<.0005
(Not checked)	39.3	975	39.7	60.3	
Other	5.2	128	53.9	46.1	.234
(Not checked)	92.9	2,306	59.6	40.4	
Do you have siblings who currently live with you?					
(Not checked)	68.7	1,707	56.5	43.5	
Yes, older one(s)	26.7	664	51.7	48.3	<.0005
(Not checked)	71.0	1,763	62.6	37.4	
Yes, younger one(s)	50.0	1,241	58.7	41.3	.368
(Not checked)	47.8	1,186	60.5	39.5	
(Same age, written as comment)	0.4	10	30.0	70.0	.112
(Not written as comment)	97.3	2,417	59.7	40.3	

NOTE: n = number; permit = learner's permit; GIS = geographic information system.

^aSome totals do not equal 100% because some students did not answer all questions.^bBased on chi-square statistic; values <.05 in bold.

TABLE 5 High School Seniors Ages 16½ and Older With and Without a License or Learner's Permit by Demographic Characteristic

Characteristic	Share of Students ^a (%)	n	Obtain License (%)		No License (%)	p-Value ^b
			Before Age 16½ Years	At or After Age 16½ Years		
Seniors age ≥ 16½						
Davis	50.0	365	30.1	37.3	32.6	<.0005
Sequoia	29.9	218	14.7	29.8	55.5	
Tamalpais	20.1	147	50.3	25.2	24.5	
Age group (years)						
16.50–16.99	0.7	5	20.0	20.0	60.0	.003
17.00–17.49	5.3	39	20.5	30.8	48.7	
17.50–17.99	34.7	253	34.8	28.9	36.4	
18.00–18.49	48.4	353	30.6	36.0	33.4	
≥18.50	11.0	80	13.8	31.3	55.0	
What is your gender?						
Male	43.4	317	25.9	39.1	35.0	.004
Female	54.5	398	32.4	27.4	40.2	
What is the highest level of education completed by whichever parent–guardian has the most education?						
Other or do not know	9.2	67	9.0	13.4	77.6	<.0005
Less than bachelor's degree	25.3	185	13.0	31.4	55.7	
Bachelor's degree or higher	63.2	461	39.9	35.6	24.5	
Do you live with your parents or guardians in one home, or split your time at different homes of separated parents?						
I live in one place.	85.5	624	28.8	31.6	39.6	.003
I live at more than one place.	11.9	87	37.9	41.4	20.7	
Distance from home to school (GIS calculated) (mi)						
0–0.49	4.7	34	32.4	23.5	44.1	.181
0.50–0.99	14.1	103	20.4	38.8	40.8	
1.00–1.99	27.3	199	36.7	31.2	32.2	
2.00–3.99	22.5	164	31.1	34.1	34.8	
≥4.00	11.2	82	34.1	36.6	29.3	
How would you describe your race–ethnicity? (Check all that apply.)						
American Indian or Native American	2.7	20	30.0	25.0	45.0	.728
(Not checked)	94.9	693	29.9	32.6	37.5	
Asian	13.3	97	19.6	38.1	42.3	.057
(Not checked)	84.4	616	31.5	31.5	37.0	
Black or African American	4.2	31	19.4	19.4	61.3	.022
(Not checked)	93.4	682	30.4	33.0	36.7	
Hispanic or Latino	27.4	200	7.5	33.5	59.0	<.0005
(Not checked)	70.3	513	38.6	32.0	29.4	
Pacific Islander	4.1	30	6.7	26.7	66.7	.002
(Not checked)	93.6	683	30.9	32.7	36.5	
White or Caucasian	60.5	442	41.9	35.7	22.4	<.0005
(Not checked)	37.1	271	10.3	26.9	62.7	
Other	4.1	30	33.3	36.7	30.0	.671
(Not checked)	93.6	683	29.7	32.2	38.1	
Do you have siblings who currently live with you?						
No	33.3	243	42.0	28.4	29.6	<.0005
(Not checked)	63.7	465	23.7	35.3	41.1	
Yes, older one(s)	21.1	154	20.1	31.2	48.7	.001
(Not checked)	75.9	554	32.7	33.4	33.9	
Yes, younger one(s)	48.9	357	24.9	35.9	39.2	.012
(Not checked)	48.1	351	35.0	29.9	35.0	
(Same age, written as comment)	0.4	3	0.0	66.7	33.3	.376
(Not written as comment)	96.6	705	30.1	32.8	37.2	

^aSome totals do not equal 100% because some students did not answer all questions.^bBased on chi-square statistic; values <.05 in bold.

holding. Students who described themselves as white–Caucasian were significantly more likely to have licenses and obtained them earlier than other students, while students who described themselves as black–African American, Hispanic–Latino, or Pacific Islander were less likely to have licenses and obtained them later than other students. Gender was not associated with differences in license holding for all students, but senior males reported having licenses at a higher rate but obtaining them later than senior females. Students who reported splitting their time between two households were also more likely to have driver’s licenses and obtain them earlier than those who reported living in only one household. Students with older or younger siblings also had a lower rate of license holding and obtained their licenses later than students who did not have siblings. No significant association was observed between the distance from home to school and license holding.

It was observed that activities and behaviors differed between those with licenses and those without licenses (Tables 6 and 7). Students who frequently participated in after-school activities, whether at school or elsewhere, or who arrived before or after the normal school start time, were more likely to have licenses and to obtain their licenses

earlier than those who did not. Students who owned a smartphone or who used electronic communications to arrange transportation were also more likely to have a license than those who did not. Students who owned bicycles were more likely to have licenses and more likely to obtain their licenses at an earlier age than those who did not own bicycles. However, those students who rode their bicycles less frequently were most likely to have licenses and to obtain licenses at an earlier age than those who rode their bicycles more frequently. Being driven in a personal vehicle while in middle school had no apparent association with licensing behavior.

A slightly greater number of associations was found between attitudes and license holding for the larger student population (Table 8) than for seniors (Table 9), likely because of the significance of grade level (shown in Table 4). Students with licenses or who obtained their licenses earlier than others were more likely to report being rushed in the morning, have lots of belongings to carry to school, and to like being physically active. Additionally, students with licenses and who obtained licenses earlier than others were more likely to report being comfortable getting to places on their own and being allowed by their parents to go places on their own. Students who

TABLE 6 Students Ages 15½ and Older With and Without a License or Learner’s Permit by Activity and Behavior Factor

Variable	Share of Students ^a (%)	<i>n</i>	License or Permit (%)		<i>p</i> -Value ^b
			Have	Do Not Have	
How many days in the school week do you currently participate in after-school activities at school?					
Rarely or never to 2	63.7	1,581	54.1	45.9	<.0005
3 to 5	35.0	869	68.8	31.2	
How many days in the school week do you currently participate in after-school activities somewhere else?					
Rarely or never to 2	65.1	1,617	54.2	45.8	<.0005
3 to 5	34.2	849	69.7	30.3	
When do you usually arrive at school?					
For activities before 1st period or after 1st period ^c	78.4	1,947	60.8	39.2	.016
For 1st period	21.0	522	54.8	45.2	
Do you have a cell phone?					
Yes, smartphone with data plan	72.0	1,787	66.7	33.3	<.0005
Yes, not a smartphone	21.5	534	45.3	54.7	
No	4.8	120	19.2	80.8	
How often do you use a cell phone, texting, e-mail, instant messaging, or other electronic communications to arrange transportation with someone? (Examples: find a ride, arrange to take the bus with a friend, tell your parents about a change in plans related to transportation, etc.) (Check one.)					
Once per week or never	72.5	1,801	62.1	37.9	<.0005
Every day, most, or a few days per week	27.1	674	52.5	47.5	
Do you currently own or have regular access to a functioning bicycle?					
Yes	76.2	1,891	64.3	35.7	<.0005
No	23.6	585	44.1	55.9	
How often do you ride your bicycle to places other than school? (Check one.)					
Never	46.9	1,165	59.0	41.0	<.0005
Once per week or less	26.5	658	67.8	32.2	
A few days per week	13.4	333	58.6	41.4	
Most days of the week	7.0	173	49.1	50.9	
Every day	5.4	133	39.1	60.9	
How did you usually get to middle school? (Check one.)					
A friend, family member, or another parent drove me.	42.8	1,062	58.9	41.1	.464
Other	55.7	1,382	60.5	39.5	

^aSome totals do not equal 100% because some students did not answer all questions.

^bBased on chi-square statistic; values <.05 in bold.

^cWording adjusted for each school to accommodate that school’s conventions for naming and numbering period.

TABLE 7 High School Seniors Ages 16½ and Older With and Without License or Learner's Permit by Activity and Behavior Factor

Variable	Share of Students ^a (%)	n	Obtain License (%)		No License or Permit (%)	p-Value ^b
			Before Age 16½ Years	At or After Age 16½ Years		
How many days in the school week do you currently participate in after-school activities at school?						
Rarely or never to 2	67.8	495	24.4	31.9	43.6	<.0005
3 to 5	31.2	228	40.4	33.8	25.9	
How many days in the school week do you currently participate in after-school activities somewhere else?						
Rarely or never to 2	64.1	468	25.2	29.9	44.9	<.0005
3 to 5	35.3	258	36.8	38.0	25.2	
When do you usually arrive at school? ^c						
For activities before 1st period or after 1st period	74.0	540	31.5	33.3	35.2	.045
For 1st period	25.2	184	24.5	30.4	45.1	
Do you have a cell phone?						
Yes, smartphone with data plan	77.3	564	33.9	35.3	30.9	<.0005
Yes, not a smartphone	18.4	134	15.7	28.4	56.0	
No	3.0	22	4.5	0.0	95.5	
How often do you use a cell phone, texting, e-mail, instant messaging, or other electronic communications to arrange transportation with someone? (Examples: find a ride, arrange to take the bus with a friend, tell your parents about a change in plans related to transportation, etc.) (Check one.)						
Once per week or never	72.6	530	30.6	32.3	37.2	.664
Every day, most, or a few days per week	27.3	199	27.1	33.7	39.2	
Do you currently own or have regular access to a functioning bicycle?						
Yes	74.2	542	32.8	35.1	32.1	<.0005
No	25.5	186	20.4	25.8	53.8	
How often do you ride your bicycle to places other than school? (Check one.)						
Never	54.1	395	33.7	29.4	37.0	<.0005
Once per week or less	25.9	189	30.7	39.2	30.2	
A few days per week	9.3	68	19.1	39.7	41.2	
Most days of the week	6.3	46	21.7	28.3	50.0	
Every day	4.0	29	3.4	24.1	72.4	
How did you usually get to middle school? (Check one.)						
A friend, family member, or another parent drove me.	39.0	285	29.1	31.6	39.3	.601
Other	58.9	430	30.7	33.7	35.6	

^aSome totals do not equal 100% because some students did not answer all questions.

^bBased on chi-square statistic; values <.05 in bold.

^cWording adjusted for each school to accommodate that school's period naming and numbering conventions.

had licenses or obtained them early were less likely to like riding the bus or reported that they could rely on their parents or guardians to drive them places. Students who had licenses or obtained them early were more likely to consider the traffic encountered going to or returning from school to be a major annoyance, consider it hilly between their homes and school, and consider going to and returning from school with friends to be important. Interestingly, students who had licenses or obtained them early were more likely to say that lots of people bicycled in their community and that their parents bicycled frequently, but less likely to say that their friends bicycled to school. For all students, those who had licenses were more likely to say that driving was the coolest way to get to school and that there was a safe bicycle route to school, but there was no significant difference in this area for seniors.

Notably, no significant association was found between licensing and students who liked being driven, liked bicycling, or considered protecting the environment important.

Models

Models were created for license holding incorporating all hypothesized variables for all students of ages 15½ and older (Table 10) and age of licensing for seniors of ages 16½ and older (Table 11).

McFadden pseudo R^2 for the binomial logit model of license holding by all students of ages 15½ and older was .50. In this model, significant demographic variables were dummy variables for white–Caucasian and Asian race–ethnicity and a parental education level of bachelor's degree or higher. Control variables of age and grade were similar in the strength of their relationship to license holding. It is notable that although these variables are highly correlated ($\rho = .82$), they are both significant, suggesting that grade may have an effect above and beyond the effect of age. Of the activity variables, smartphone ownership had the largest effect. Participation in activities after school, bicycle access and use, and school arrival time also had significant effects. Of the attitude variables, liking riding

TABLE 8 Students Ages 15½ and Older Who Agree or Strongly Agree to Attitudinal Statements

Statement	Share of All Students (%)	License or Permit (%)		<i>p</i> -Value ^a
		Have	Do not Have	
I am always rushed to get ready in the morning.	50.3	53.7	46.8	.001
I have lots of stuff to carry to school.	49.2	55.1	41.8	<.0005
I like being physically active.	80.4	85.8	74.2	<.0005
I like bicycling.	51.6	51.5	54.0	.250
I like riding the bus.	17.2	14.7	21.3	<.0005
I like being driven places.	56.1	57.3	56.0	.570
Driving is considered the coolest way to get to school.	48.6	52.5	44.9	<.0005
Bicycling is considered the coolest way to get to school.	4.6	4.3	5.4	.242
Going to and from school with friends rather than alone is a priority.	16.8	15.2	19.7	.004
Protecting the environment is important to me.	63.4	64.2	63.7	.826
The traffic congestion getting in and out of school is a major hassle.	44.1	50.3	36.8	<.0005
I can rely on my parents or guardians to drive me places.	55.5	54.0	58.8	.021
Lots of people bicycle in my community.	69.2	74.5	63.9	<.0005
My parents or guardians encourage me to bicycle.	30.2	30.2	30.8	.780
I feel comfortable getting places on my own.	81.2	88.7	72.1	<.0005
My parents or guardians allow me to go places on my own.	82.6	91.4	71.6	<.0005
One or both of my parents or guardians bicycle frequently.	18.8	21.1	16.2	.003
There is a safe route to bicycle from my home to school.	56.9	62.2	50.9	<.0005
My friends bicycle to school.	31.1	27.9	37.0	<.0005
I live too far away from school to bicycle there.	25.6	26.7	24.9	.357
It is hilly between my home and school.	26.8	31.1	21.6	<.0005

NOTE: *N* = 2,483. Answers were scored on five-point scale from strongly disagree (1) to agree strongly (5); percentages of all students are based on the sample of students who answered the corresponding question; sizes ranged from 2,439 to 2,464, with a mean of 2,451.

The item for which most data was missing was "Bicycling is considered the coolest way to get to school"; most others were missing no more than 33 responses.

^aBased on chi-square statistic; values <.05 in bold.

TABLE 9 High School Seniors Ages 16½ and Older Who Agree or Strongly Agree to Attitudinal Statements

Characteristic	Share of All Students (%)	Students With License (%)		No License (%)	<i>p</i> -Value ^a
		Before Age 16½ Years	At or After Age 16½ Years		
I am always rushed to get ready in the morning.	55.1	62.5	53.8	51.9	.050
I have lots of stuff to carry to school.	48.6	61.1	50.9	38.4	<.0005
I like being physically active.	81.0	88.9	83.9	73.9	<.0005
I like bicycling.	51.1	50.9	53.0	51.7	.906
I like riding the bus.	16.2	11.1	14.8	21.6	.006
I like being driven places.	52.6	54.6	49.8	54.8	.457
Driving is considered the coolest way to get to school.	45.2	44.4	48.7	44.6	.580
Bicycling is considered the coolest way to get to school.	5.2	4.2	4.3	7.0	.271
Going to and from school with friends rather than alone is a priority.	12.2	9.3	12.3	14.8	.185
Protecting the environment is important to me.	62.9	61.6	63.3	64.2	.831
The traffic congestion getting in and out of school is a major hassle.	49.7	60.2	54.9	38.7	<.0005
I can rely on my parents or guardians to drive me places.	48.9	41.7	47.3	56.8	.003
Lots of people bicycle in my community.	71.1	82.4	72.5	63.9	<.0005
My parents or guardians encourage me to bicycle.	23.6	16.7	28.0	25.5	.013
I feel comfortable getting places on my own.	85.9	94.0	93.7	74.1	<.0005
My parents or guardians allow me to go places on my own.	86.6	98.1	95.4	72.0	<.0005
One or both of my parents or guardians bicycle frequently.	17.0	23.3	16.2	13.2	.012
There is a safe route to bicycle from my home to school.	55.8	59.3	59.1	52.0	.173
My friends bicycle to school.	21.8	16.4	18.8	29.6	.001
I live too far away from school to bicycle there.	25.2	29.3	25.6	22.6	.243
It is hilly between my home and school.	28.5	35.8	29.3	20.2	.001

NOTE: *N* = 730. Answers scored on five-point scale from strongly disagree (1) to agree strongly (5); shares of all students are based on the sample who answered the corresponding question; sizes ranged from 718 to 727, with a mean of 722. The items for which most data was missing were "Lots of people bicycle in my community" and "My friends bicycle to school"; most others were missing no more than eight responses.

^aBased on chi-square statistic; values <.05 in bold.

TABLE 10 Model of Having a License or Learner's Permit Versus Not Having a License or Learner's Permit for Students Ages 15½ and Older, by Demographic Characteristic

Variable	Coefficients	SE	z-Value	Pr(> z) ^a	Odds Ratio
Constant	-15.550	1.981	-7.850	<.0005	0.00
Demographics					
White or Caucasian	0.879	0.176	4.980	<.0005	2.41
Parent has at least bachelor's degree	0.758	0.172	4.422	<.0005	2.13
Grade	0.650	0.173	3.762	<.0005	1.92
Asian	-0.490	0.202	-2.425	.015	0.61
Age	0.337	0.167	2.016	.044	1.40
Attend Sequoia	-0.364	0.188	-1.939	.053	0.69
Hispanic or Latino	-0.306	0.195	-1.570	.116	0.74
Female	-0.164	0.146	-1.125	.261	0.85
Black or African American	-0.319	0.337	-0.947	.344	0.73
Increasing distance to school	0.060	0.065	0.932	.351	1.06
Attend Tamalpais	0.176	0.198	0.886	.376	1.19
Split time between homes	0.195	0.233	0.835	.404	1.21
Has older siblings	0.092	0.160	0.578	.563	1.10
Has younger siblings	0.034	0.139	0.246	.806	1.03
Activity factors					
Own a smartphone	0.972	0.154	6.314	<.0005	2.64
Ride bicycle to places other than school most days or every day per week	-1.068	0.204	-5.249	<.0005	0.34
Participate in after school activities at school 3 to 5 days per week	0.763	0.153	4.991	<.0005	2.14
Participate in after school activities elsewhere 3 to 5 days per week	0.463	0.148	3.139	.002	1.59
Arrive before or after regular school start time	-0.517	0.166	-3.106	.002	0.60
Regular access to a functioning bicycle	0.405	0.184	2.195	.028	1.50
Use cell phone to arrange transportation	0.243	0.156	1.557	.119	1.27
Driven in personal vehicle to middle school	-0.087	0.149	-0.584	.560	0.92
Attitudinal statements					
Like riding the bus	-0.549	0.175	-3.131	.002	0.58
Can rely on parents or guardians to drive places	-0.317	0.143	-2.218	.027	0.73
Lots of stuff to carry	0.296	0.144	2.052	.040	1.35
Consider driving coolest way to get to school	0.174	0.139	1.250	.211	1.19
Like being physically active	0.216	0.197	1.096	.273	1.24
Like biking	-0.115	0.147	-0.777	.437	0.89
Protecting the environment important	-0.091	0.155	-0.588	.556	0.91
Consider biking coolest way to get to school	-0.163	0.351	-0.465	.642	0.85
Always rushed in the morning	-0.016	0.138	-0.118	.906	0.98
Going to and from school with friends priority	-0.011	0.186	-0.058	.954	0.99
Like being driven	0.000	0.142	0.003	.998	1.00

NOTE: SE = standard error; valid $N = 1,462$; McFadden pseudo- $R^2 = .50$; log likelihood (full model) = -699.04; and log likelihood (constant only model) = -1,393.71.

^aValues <.05 in bold.

the bus and ability to rely on parents for driving were negatively related, while having a lot to carry was positively related to license holding.

McFadden pseudo R^2 for the multinomial logit model of age of license acquisition by all high school seniors of ages 16½ and older was .30. Variables significantly related to not holding a license were similar to those in the model of license holding of all students of ages 15½ and older. Additionally, students who liked being physically active were less likely to have a license than others. However, fewer variables were related to the age of licensing for high school seniors. The strongest relationships were with Hispanic-Latino ethnicity, having younger siblings, and age, which were all related to later licensing. Female students, Tamalpais High students, students with smartphones, and students who liked being physically active were more likely to obtain a license early, while those who liked to ride the bus were less likely to do so.

Results of both the models and the bivariate analysis were generally consistent with the hypotheses, though there were some exceptions. Demographically, gender had less of an association with licensing

overall; of the high school seniors, female students were more likely to obtain their licenses early, but also less likely overall to have a license. Distance from home to school had no significant association, contrary to expectations. Asian ethnicity was associated with less and later licensing, also contrary to expectations, which were based on the association of Asian ethnicity with a higher income (22). Having younger siblings had no correlation with having a license, but high school seniors without younger siblings were likely to obtain their licenses early. Of the activities and behaviors surveyed, cell phone or smartphone ownership was associated with greater and earlier licensing. Bicycle ownership was associated with increased and earlier license holding, contrary to frequency of bicycling (liking bicycling had no significant association with license holding). Additionally, having been driven to middle school had no association with licensing. Similarly, several attitudinal factors had relationships with licensing contrary to expectations. Liking physical activity had a positive correlation with licensing. Importance of protecting the environment had no correlation with licensing, nor did considering bicycling cool or liking being driven.

TABLE 11 Model of Licensing Age Versus Demographic Characteristic for High School Seniors Ages 16½ and Older

Variable	License Late					No License				
	Coefficient	SE	t-Value	Pr(> t) ^a	Relative Risk Ratio	Coefficient	SE	t-Value	Pr(> t) ^a	Relative Risk Ratio
Constant	-20.685	7.338	-2.819	.005	0.00	-0.337	8.115	-0.042	.967	0.71
Demographics										
Hispanic or Latino	1.771	0.456	3.880	<.0005	5.88	1.281	0.527	2.432	.015	3.60
Has younger siblings	0.781	0.263	2.970	.003	2.18	0.449	0.318	1.411	.158	1.57
Split time between homes	0.535	0.359	1.491	.136	1.71	-1.128	0.601	-1.876	.061	0.32
Attend Tamalpais	-0.870	0.347	-2.506	.012	0.42	-0.457	0.452	-1.010	.312	0.63
White or Caucasian	-0.406	0.417	-0.973	.330	0.67	-1.921	0.461	-4.171	<.0005	0.15
Age	1.110	0.383	2.898	.004	3.04	0.343	0.426	0.805	.421	1.41
Parent has at least bachelor's degree	-0.189	0.303	-0.623	.534	0.83	-0.927	0.311	-2.978	.003	0.40
Female	-0.776	0.296	-2.618	.009	0.46	0.152	0.352	0.431	.666	1.16
Asian	0.566	0.445	1.271	.204	1.76	0.344	0.520	0.662	.508	1.41
Attend Sequoia	0.210	0.389	0.539	.590	1.23	0.498	0.458	1.089	.276	1.65
Has older siblings	0.377	0.354	1.065	.287	1.46	0.146	0.408	0.358	.720	1.16
Increasing distance to school	-0.087	0.129	-0.676	.499	0.92	-0.069	0.151	-0.455	.649	0.93
Black or African American	0.103	0.980	0.105	.917	1.11	0.655	0.990	0.661	.509	1.92
Activity factors										
Own a smartphone	-0.794	0.380	-2.090	.037	0.45	-2.078	0.409	-5.086	<.0005	0.13
Ride bicycle to places other than school most days or every day per week	0.916	0.502	1.826	.068	2.50	2.374	0.528	4.498	<.0005	10.74
Arrive before or after regular school start time	0.362	0.296	1.222	.222	1.44	0.526	0.346	1.518	.129	1.69
Participate in after school activities elsewhere 3 to 5 days per week	-0.182	0.261	-0.697	.486	0.83	-1.050	0.335	-3.135	.002	0.35
Participate in after school activities at school 3 to 5 days per week	-0.170	0.274	-0.618	.536	0.84	-0.896	0.346	-2.588	.010	0.41
Regular access to a functioning bicycle	-0.101	0.353	-0.286	.775	0.90	-0.280	0.395	-0.709	.478	0.76
Driven in personal vehicle to middle school	0.211	0.299	0.706	.480	1.24	-0.080	0.358	-0.224	.823	0.92
Use cell phone to arrange transportation	-0.040	0.291	-0.137	.891	0.96	-0.064	0.341	-0.186	.852	0.94
Attitudinal statements										
Like riding the bus	1.086	0.396	2.744	.006	2.96	1.409	0.444	3.172	.002	4.09
Like being physically active	-0.893	0.407	-2.191	.028	0.41	-1.160	0.460	-2.525	.012	0.31
Always rushed in the morning	-0.466	0.267	-1.744	.081	0.63	-0.918	0.326	-2.819	.005	0.40
Can rely on parents or guardians to drive places	0.426	0.262	1.627	.104	1.53	0.935	0.315	2.964	.003	2.55
Protecting the environment important	0.535	0.296	1.809	.071	1.71	0.582	0.362	1.608	.108	1.79
Lots of stuff to carry	-0.230	0.284	-0.809	.419	0.79	-0.657	0.340	-1.934	.053	0.52
Consider biking coolest way to get to school	-0.816	0.646	-1.264	.206	0.44	-0.596	0.697	-0.855	.393	0.55
Consider driving coolest way to get to school	0.378	0.260	1.452	.146	1.46	0.134	0.315	0.425	.671	1.14
Going to and from school with friends priority	0.298	0.417	0.715	.475	1.35	-0.397	0.522	-0.760	.447	0.67
Like being driven	-0.501	0.267	-1.878	.060	0.61	-0.054	0.325	-0.166	.868	0.95
Like biking	-0.259	0.290	-0.893	.372	0.77	-0.137	0.346	-0.397	.692	0.87

NOTE: License late = obtain license or permit at or after age 16½ years as compared with obtain license or permit before age 16½ years; no license = no license or permit as compared with obtain license or permit before age 16½ years; valid $N = 518$; McFadden pseudo- $R^2 = .30$; log likelihood (full model) = -398.92; and log likelihood (constant-only model) = -569.00.

^aValues <.05 in bold.

DISCUSSION AND CONCLUSIONS

Results of this analysis provide new insights into factors influencing delayed licensing. Similar to studies by Williams (9) and Tefft et al. (5), this study found strong relationships between licensing and demographic factors. In particular, greater licensing rates were observed for white–Caucasian students and later licensing for black and Hispanic respondents. Though no direct question about income was included in the survey, increased licensing rates and earlier licensing were observed for variables that may correspond to income, such as higher parental education levels, smartphone ownership, and bicycle ownership. Though these findings should not necessarily be generalized beyond this sample, this result is consistent with findings by Williams (9), Tefft et al. (5), Curry et al. (11), and Schoettle and Sivak (4).

The analysis also points to the importance of participation in activities and selected behaviors in explaining delayed licensing. Though use of electronic communication methods for arranging transportation was associated with higher licensing rates and earlier licensing in bivariate analysis, this factor was not significant in either model. This result is consistent with the observations made about the use of electronic communications by Williams (9), Tefft et al. (5), and Le Vine et al. (16). An association was also observed between participation in after-school activities and higher rates of licensing and earlier licensing. This observation appears to contradict the finding that students' busyness is a reason for lower licensing rates and later licensing, as self-reported by respondents to surveys by Williams (9) and Tefft et al. (5). In the present study, students who participated frequently in after-school activities were more likely to have licenses and to obtain those licenses earlier than students who participated in such activities less frequently. However, this result does not necessarily contradict a conclusion that an increase in student activities may be a contributing factor to a decrease in licensing over time. The differences between similar students over time are not necessarily the same as differences between various students at one time. In addition, the causal link between licensing and participation in after-school activities may work in both directions, that is, licensing may affect participation and participation may affect licensing, both in terms of the rate of licensing and the age at which students first obtain a license.

This analysis indicated that attitudes are also important. The finding of the current study that environmental concerns are apparently not a factor in licensing is consistent with the findings of Schoettle and Sivak (4) and Le Vine et al. (18). However, the associations observed for this study between enjoyment of bicycling and the perception of bicycling in the community (as well as bicycle use) with decreased licensing, as well as enjoyment of riding the bus with decreased licensing, also support the conclusion that may be drawn from Schoettle and Sivak that students do not need a license to do what they want (4) and the conclusion from Davis et al. that young adults increasingly use transportation modes other than driving (12). Positive attitudes about other transportation modes or the availability of transportation provided by parents may be correlated with students who do not need to drive to get to their desired locations.

These results indicate that trends in licensing may be more related to broader societal factors than to specific policies. The strong relationships with race–ethnicity, variables suspected to be related to income (including parental education, bicycle ownership, and smartphone ownership), and participation in after-school activities imply that larger trends may underlie current behaviors more than do specific transportation policies. Some factors, such as attitudes about the environment,

bicycle ownership, or smartphone ownership, may have no effect or an effect opposite to what has generally been expected. However, this analysis also points to areas for which specific policy interventions may have an impact on driving by younger drivers. The association with bicycle use and attitudes about riding the bus and availability of parental transportation implies that policies influencing these factors may have an effect on licensing rates.

Additional research may help clarify these trends and policy implications. Further work is needed to better separate the effects of income from other factors such as electronic communications, bicycle ownership, smartphone ownership, and after-school activities. Although there was an attempt to investigate these effects in this study through a survey of different schools, direct measurements of income as well as a wider survey with a greater range of transportation networks, city sizes, land use patterns, and built environment characteristics would also help clarify these effects as well as identify other influences on licensing. Additionally, repetition of the survey over time would support analysis and understanding of changes in licensing.

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